Milestone 01 — Download and Prepare Covariates

Structured Acquisition and QA of Inputs for Regional Skew Estimation

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Contents

1 Overview of v1.4

This document outlines the setup, documentation, and reproducibility scaffolding established for **Milestone v1.4**, focused on acquiring and validating spatial covariates for regional skew modeling.

This milestone builds on `v0.3-structure-refactor`.

1.0.1 Summary

Initiate acquisition, validation, and preparation of climate, terrain, and location-based covariates for use in regional skew estimation models.

1.1 Goals

Update the covariate source inventory:

- Write or refactor download scripts
- Document file sources and formats
  `covariate_source_inventory.md`
- Prepare spatial data
  `spatial_data_preparation_checklist.md`
- Acquire and clean metadata
  `metadata`
- Apply version control and tagging

2 Project Structure

```
FFA_regional-skew/
  .gitignore                    # Prevents sensitive/local files from being pushed
  arcgis_project/              # ArcGIS Pro .aprx project and supporting layers
  CHANGELOG.Rmd                # Human-readable changelog (semantic versioning)
  data/                        # Raw, processed, meta, intermediate, and QA data
  docs/                        # Documentation, workflow notes, metadata, checklists
  FFA_regional-skew.Rproj      # RStudio project file (keep in root)
  notebooks/                   # Ad hoc .Rmd/.qmd experiments (empty or minimal)
  notes/                       # Team notes, meeting logs, brainstorms (convert to .md/.qmd)
  R/                           # Analysis scripts (milestone-organized)
      01_download/            # Download raw data
          _latex_preamble.tex                # LaTeX preamble for PDF reports
          01a_download_ecoregions.R          # Download U.S. ecoregions
          01b_download_USGS_gage_data.R      # Download raw peak flow site data
          01c_download_usgs_gage_metadata.R  # Download USGS site metadata
          01d_download_usgs_peakflow_data.R  # Download peak flow data
          01e_filter_usgs_peakflow_data.R    # Filter peak flow data
          01f_download_nhdplus_v21_flowlines.R
          01g_download_nhdplus_hr_catchments.R
          01h_download_nhdplus_hr_flowlines.R
          01j_download_koppen_geiger_climate.R
          01k_download_plant_hardiness_zone_map.R
          01l_download_prism_climate.R
          01m_download_nlcd_2016.R
          01n_download_ned.R
          01o_download_modis_2016.R
          01p_download_statsgo2.R
      02_clean_validate/      # QA scripts and validation checks
      03_covariates/          # Covariate extraction: climate, terrain, land cover
      04_modeling/            # Statistical modeling (GAMs, Elastic Net, etc.)
      06_eval/                # Model diagnostics, residuals, validation
      utils/                  # Reusable functions (organized by domain)
          metadata/
          spatial/
          qaqc/
```

```
        paths/
        plotting/
  README.md                   # GitHub-readable overview and navigation aid
  README.Rmd                  # Editable RMarkdown version with richer formatting
  reports/                    # Knitted .Rmd/.qmd milestone reports
                               # Next: consider `reports/final/` and `reports/draft/`

  results/                    # Manuscript-ready figures, models, and outputs
    maps/
    tables/
    models/
    posterdown/              # Poster files and assets
    slides/                  # Slide decks and presentation visuals
  sandbox/                    # Staging area for review
```

3 v1.4 – Download and Prepare Covariates

4 v1.4 Tasklist

Step	Task	Status
1.4.1	Refine the Covariate Inventory	[X]
1.4.2	Update folder structure	[X]
1.4.3	Create / update download scripts for vector data	[X]
1.4.4	Create downloads script for raster covariates	[X]
1.4.5	QAQC for downloads	[X]
1.4.6	Knit milestone and data dictionary .Rmd files to PDF	[ ]
1.4.7	Document changes in 01_download README.Rmd	[ ]
1.4.8	Commit and tag `v1.4-download-scripts`	[ ]

5 Changelog Format: Per-Step Narrative

5.0.1 Step 1.4.1 — Refine the Covariate Inventory

Actions:

- Finalized covariate list by domain (climate, terrain, land cover)
- Documented filenames, source URLs, formats, versioning, and priority
- Created reusable QA checklists and source tracking templates

Reason (Before):

The project lacked a centralized, versioned inventory of covariates. Metadata was fragmented across exploratory scripts with no QA framework.

Result (After):

- Created `docs/covariate_source_inventory.md`
- Created `docs/spatial_data_preparation_checklist.md` for 12-step QA
- Updated `data/meta/covariates_metadata_schema.csv`
- Standardized metadata for reproducibility and audit tracking

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5.0.2 Step 1.4.2 — Update Folder Structure for `utils/` and `data/`

Actions:

- Backed up project before restructuring
- Created new subfolders under `data/`: `raw/`, `processed/`, `meta/`, `log/`, `intermediate/`
- Created new utility folders in `R/utils/` for:
  - `metadata/`
  - `spatial/`
  - `qaqc/`
  - `paths/`
  - `plotting/`

Reason (Before):

Functions were embedded in monolithic scripts or scattered, limiting reuse and debugging.

Result (After):

Utility functions are modular, testable, and purpose-scoped. Folder structure supports test-driven development and script clarity.

5.0.3 Step 1.4.3 — Create / Update Download Scripts for Vector Data

Actions:

- Updated ecoregions, USGS gage, NHDPlus, and STATSGO2 download scripts
- Harmonized all vector outputs to EPSG:5070
- STATSGO2 spatial queries now chunked and geometry-validated

Reason (Before):

Inconsistent file structures, missing scripts, and fragile STATSGO2 queries.

Result (After):

All vector datasets reproducibly downloaded, spatially filtered to Great Plains AOI, and exported to `data/raw/` and `data/processed/`. STATSGO2 joins are verified by `mukey`.

5.0.4 Step 1.4.4 — Create Download Scripts for Raster Covariates

Actions:

- Created modular download and processing scripts for:
  - Köppen-Geiger
  - USDA Plant Hardiness Zones
  - PRISM climate normals (via `{prism}`)
  - NLCD 2016
  - NED elevation and slope (via `{terra}`)
  - MODIS NDVI 2016 (via `{MODISTools}`)
- Standardized reprojection (EPSG:5070), clipping, and export

Reason (Before):

Raster layers were manually downloaded, inconsistently named, or lacked projections.

Result (After):

All rasters downloaded, cleaned, and saved to `data/processed/` with consistent filenames, formats, and CRS. Metadata written to `data/meta/`.

5.0.5 Step 1.4.5 — QAQC for Downloads

Actions:

- Validated CRS, resolution, and extent across all vector and raster layers
- Created summary metadata export including layer name, type, CRS, resolution, extent, and notes
- Tagged `.Rmd` files with `{r name, eval=}` chunk headers for reproducibility
- Documented internal workflow in:
  - `git_changelog_workflow_reference.Rmd`
  - `spatial_data_preparation_checklist.Rmd`

Reason (Before):

Unvalidated spatial layers posed a risk of alignment errors. Documentation was scattered or incomplete.

Result (After):

All layers validated for modeling compatibility. Scripts are now reproducible, documented, and version-controlled. Project directories are aligned with workflow logic and audit-ready.

5.1 Next Steps

- [ ] Knit milestone `.Rmd` and metadata summary to PDF
- [ ] Finalize edits to `01_download/README.Rmd`
- [ ] Tag the milestone:

```
git tag -a v1.4-download-scripts -m "Milestone 01: Download scripts and covariate metadata"
git push origin v1.4-download-scripts
```